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Title: Computer Network Cabling

Objective

- List common cable types used in computer networking
- Describe how UTP cables are made
- Explain how UTP cables are used in Ethernet networks
- Demonstrate the ability to make a working patch cable
- Name the two wiring standards used for wired Ethernet networks and their uses

Common Network cable types (Guided Media)

Coaxial Cable has two wires of copper. The core wire lies in the center and it is made of solid conductor. The core is enclosed in an insulating sheath. The second wire is wrapped around over the sheath and that too in turn encased by insulator sheath. This all is covered by plastic cover.

Fig 3.1 Coaxial Cable

Twisted Pair Cable is made of two plastic insulated copper wires twisted together to form a single medium. Out of these two wires, only one carries the actual signal and another is used for ground reference. The twists between wires are helpful in reducing noise (electro -magnetic interference) and crosstalk.

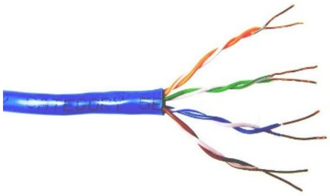

Fig 3.2 Twisted Pair Cable

Fiber Optic works on the properties of light. When light ray hits at critical angle, it tends to refract at 90 degree. This property has been used in fiber optic. The core of fiber optic cable is made of high quality glass or plastic

Fig 3.3 Fiber Optic Cable

Twisted Pair (UTP/STP) Categories

Category 1	Voice only (Telephone)
Category 2	Data to 4 Mbps (Localtalk)
Category 3	Data to 10Mbps (Ethernet)
Category 4	Data to 20Mbps (Token ring)
Category 5 Category 5e	Data to 100Mbps (Fast Ethernet) Data to 1000Mbps (Gigabit Ethernet)
Category 6	Data to 2500Mbps (Gigabit Ethernet)

Twisted Pair Characteristics

1. Unshielded or Shielded
2. Twisted pair of insulated conductor
3. Covered by insulating sheath

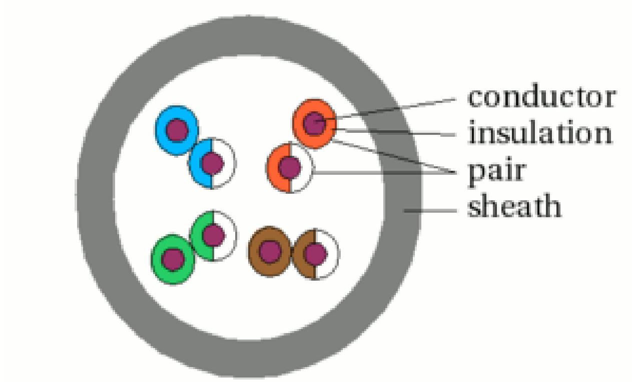

Fig 3.4 UTP/STP Cable

Task 1. Define each characteristic of copper

1. Malleability

Copper can be easily shaped without breaking. This property allows copper to be easily formed into wires and other shapes, making it ideal for use in electrical conductors.

2. Ductility

The ability of copper to be stretched into a wire without breaking. It allows copper to be drawn into thin wires that can be easily incorporated into twisted pair cables.

3.Strong

Copper withstands mechanical stress without deforming or breaking. The strength of copper ensures that the wires can endure physical handling, installation, and environmental factors.

4.Durable

Copper resists wear and damage over time. Durability ensures that the copper conductors can withstand the rigors of daily use including bending, twisting and exposure to various environmetal conditions.

5. Corrosion Resistant

Copper withstands degradation from environmental elements. These characteristics make copper an excellent choice for use in twisted pair cables, which are widely used in networking for data transmission.

Task 2. Identify Network cabling tools and equipments

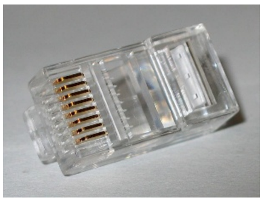

Fig 3.4 RJ45

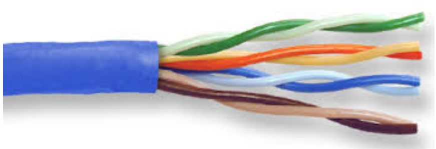

Fig 3.5 CAT5e UTP Cable

Fig 3.6 Crimping tool

Fig 3.7 Network Tester

Task 3. Creating a Straight, Crossover and Rollover connection

Step 1. Strip Cable end – avoid cutting into conductor insulation.

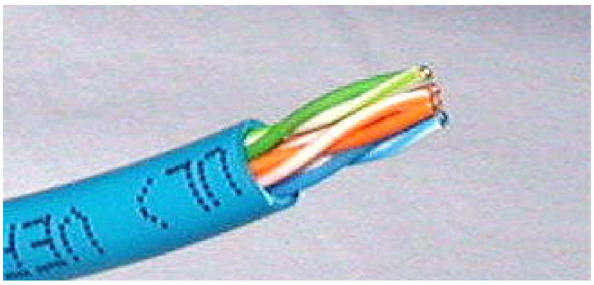

Fig 3.8 Twisted Pair Cable

Step 2. Untwist wire ends – sort wires by insulation colors

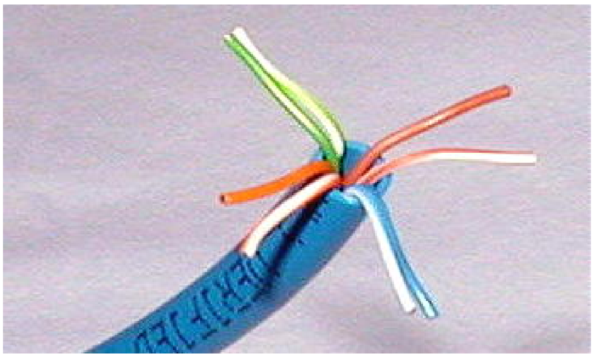

Fig 3.9 Four pairs of twisted pair Cable

Step 3. Arrange wires

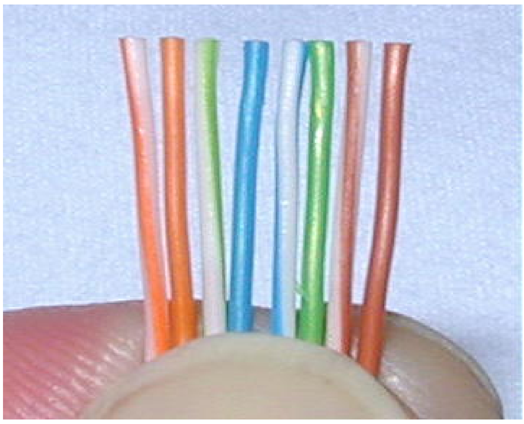

Fig 3.10 TIA/EIA 568B

Straight through Cable Connection – use to connect network device to other different network device (Example: PC to Switch, Router to Switch)

1. TIA/EIA 568A: GW-G, OW-Bl, BlW-O, BrW-Br

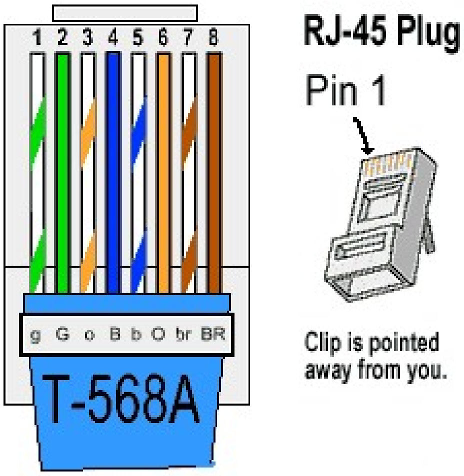

Fig 3.11 T-568A

2. TIA/EIA 568B: OW-O, GW-Bl, BlW-G, BrW-Br

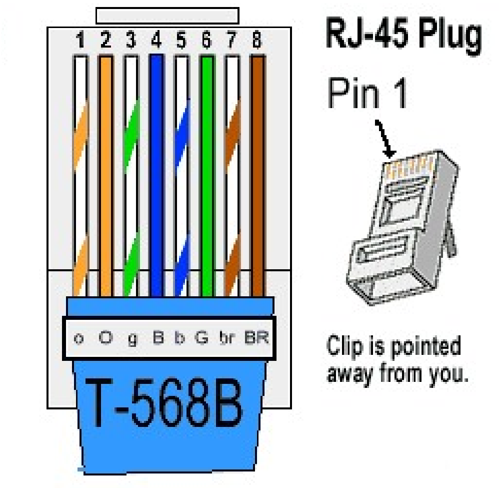

Fig 3.12 T-568B

Crossover Cable Connection – use to connect network device to other same network device (Example: PC to PC, Switch to Switch)

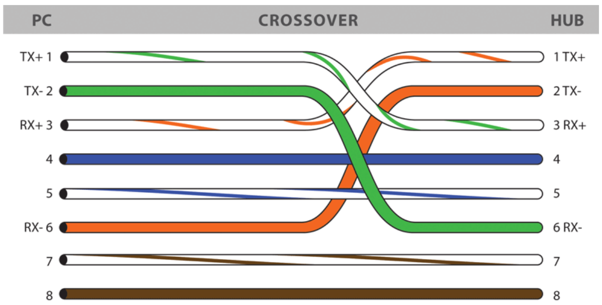

Fig 3.13 EIA/TIA 568B+568A Crossover

Rollover Cable Connection

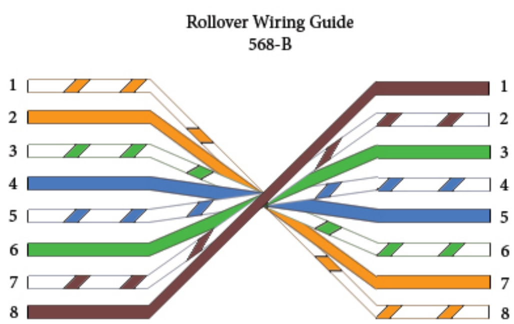

Fig 3.14 Rollover

Step 4. Trim wires to size – Trim all wires evenly, leave about ½ inch of twisted wires exposed

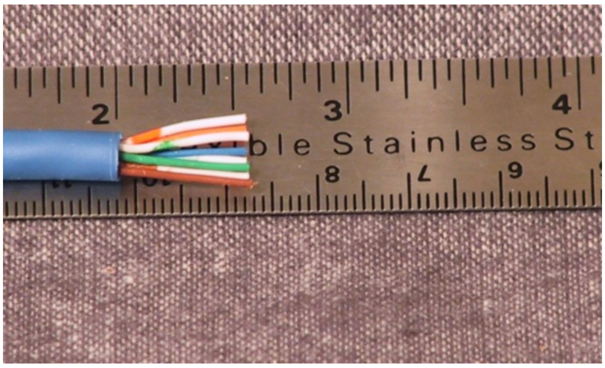

Fig 3.15 Twisted pair cable trim size

Step 5. Attach connector – Maintain wire order from left to right, with RJ45 tab facing down

Fig 3.16 Top view of RJ45 and Twisted cable connection

Step 6. Check if the cable is attached to the RJ45 correctly – check if wire extends to the end of the RJ45, check if the sheath is inserted well inside the RJ45 lock.

Fig 3.17 Side view of RJ45 and Twisted cable connection

Step 7. Crimp – Squeeze firmly to crimp connector onto cable end

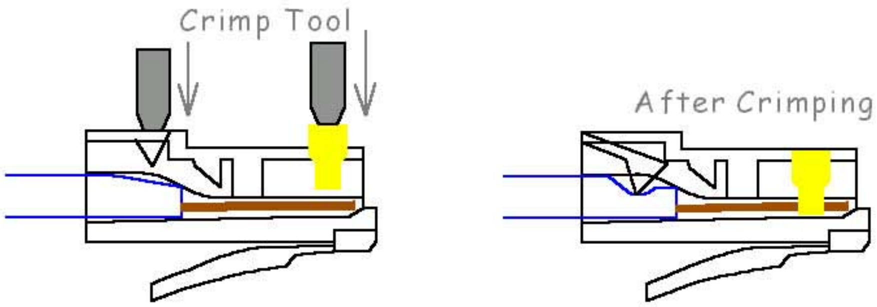

Fig 3.18 Crimping RJ45 to wire

Step 8. Test network cable – Use the network tester if the cable is working properly according to its cable connection.

Fig 3.19 Network tester